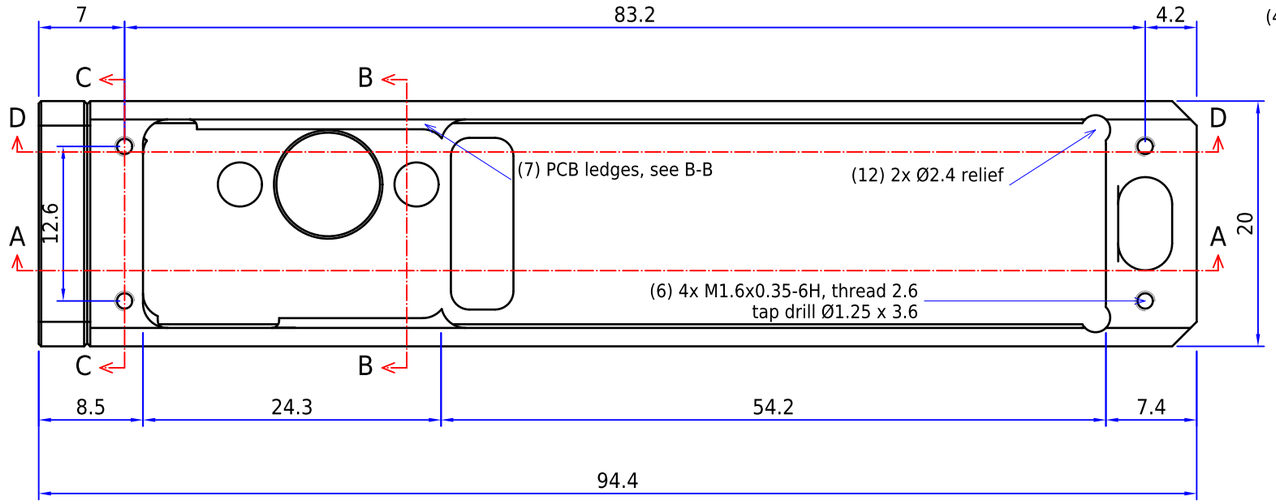
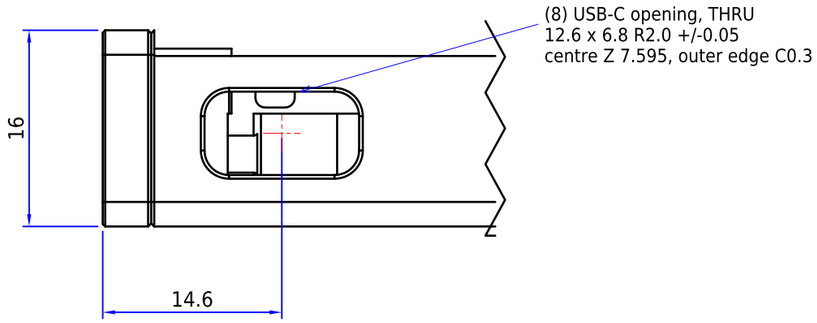
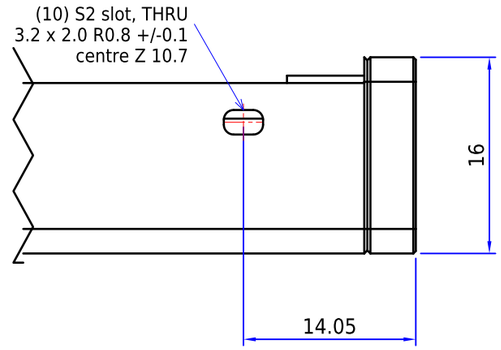


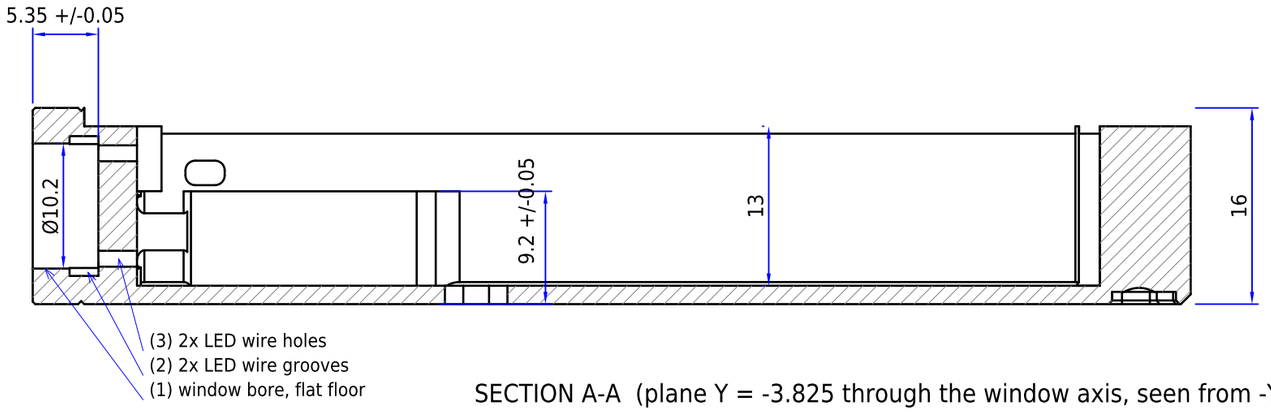
TOP VIEW (from +Z, lid removed) items 6, 7, 11, 12



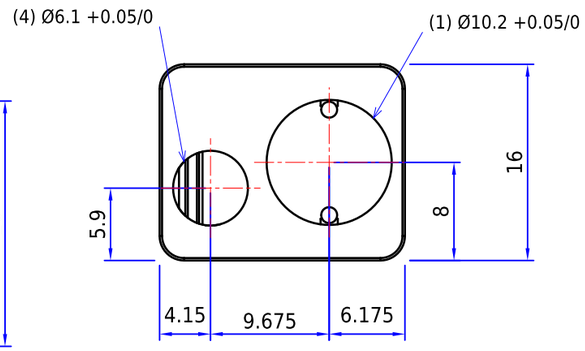
SIDE VIEW from -Y (control side) item 8



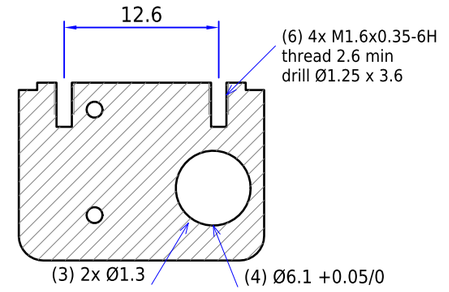
SIDE VIEW from +Y (laser side) item 10



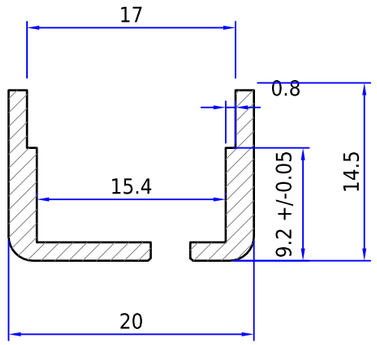
SECTION A-A (plane Y = -3.825 through the window axis, seen from -Y) items 1, 2, 3, 7, 11



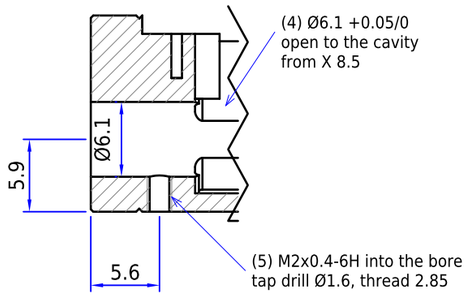
FRONT VIEW (from -X) items 1-4



SECTION C-C (X = 7) items 3, 4, 6



SECTION B-B (X = 30) items 7, 11



SECTION D-D (Y = 5.85) items 4, 5

ID	FEATURE	SPECIFICATION	TOLERANCE / NOTE
1	Window bore (front)	Ø10.2 x 5.35 deep, flat floor; axis Y -3.825, Z 8.0 = 6.175 from the control-side face	Ø +0.05/0; depth +/-0.05 floor flat 0.03, Ra 1.6
2	LED wire grooves 2x	at 12 and 6 o'clock: 1.4 wide, bottom R5.7 from the axis, X 3.0 to 5.35 (bore floor)	+/-0.1
3	LED wire holes 2x	Ø1.3 from the bore floor into the cavity (X 5.35 to 8.5), at Z 8.0 +/- 4.3, Y -3.825	+/-0.1
4	Laser bore (front)	Ø6.1, axis Y 5.85 (4.15 from the +Y face), Z 5.9; open to the cavity at X 8.5 (module 6 x 12, recess 0.3)	Ø +0.05/0; axis +/-0.05
5	Set-screw hole (bottom)	M2x0.4-6H from the bottom face into the laser bore, X 5.6, Y 5.85; tap drill Ø1.6, thread length 2.85	6H; position +/-0.1
6	Lid screw threads 4x	M1.6x0.35-6H, drill Ø1.25 x 3.6 deep (flat), thread 2.6 min; X 7.0 and 90.2, Y +/- 6.3	6H; position +/-0.1
7	PCB ledges	Z 9.2, 0.8 wide, X 8.5 to 32.8. None: -Y at X 9.7 to 19.6 (USB-C); +Y at X < 12.9 (laser)	Z +/-0.05; both ledges coplanar within 0.05
8	USB-C opening (-Y wall)	12.6 x 6.8 R2.0 through; centre X 14.6, Z 7.595; outer edge C0.3	+/-0.05
9	Floor holes (bottom face)	main Ø8.45 thru + counterbore Ø8.85 x 0.4 deep from inside; 2 slide holes Ø3.6 thru (0.2 x 45 deg chamfer); window: sheet 2	Ø +0.05/0; position +/-0.05
10	S2 slot (+Y wall)	3.2 x 2.0 R0.8 through; centre X 14.05, Z 10.7	+/-0.1
11	Cavity	17.0 wide (1.5 walls), floor Z 1.5, lid seat Z 14.5; vertical corners R2.0; trough 15.4 wide between the ledges	+/-0.1
12	Bay corner reliefs 2x	Ø2.4 through the bay-end corners, centres X 86.151, Y +/- 7.651	+/-0.1
13	Exterior, lid, marking	floor holes and window, V-ring and skin seams, tail and lanyard slot, marking: sheet 2. Lid: drawing FK-H-02.	

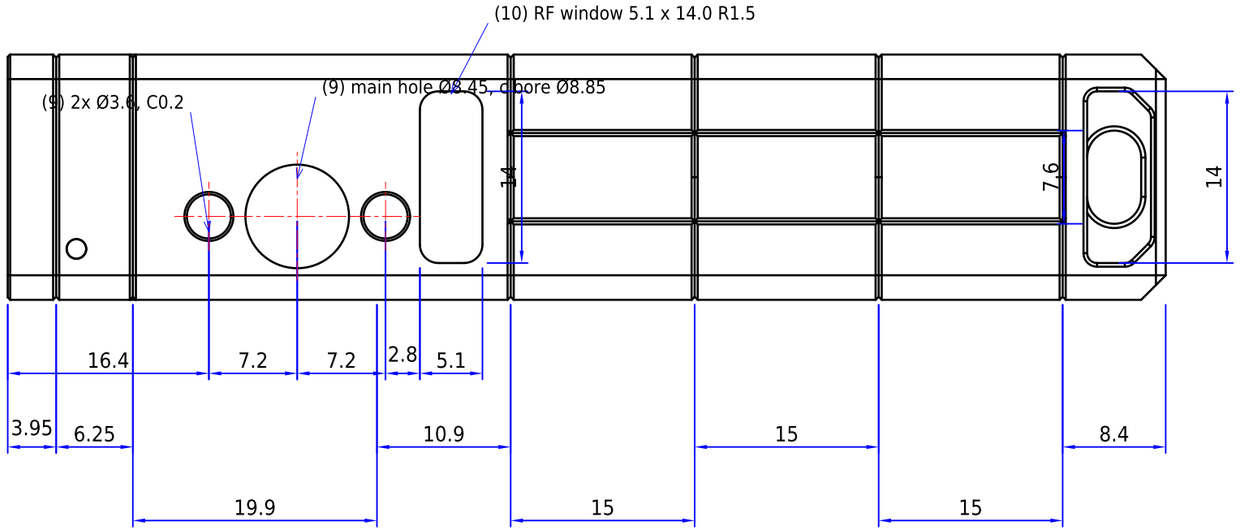
NOTES

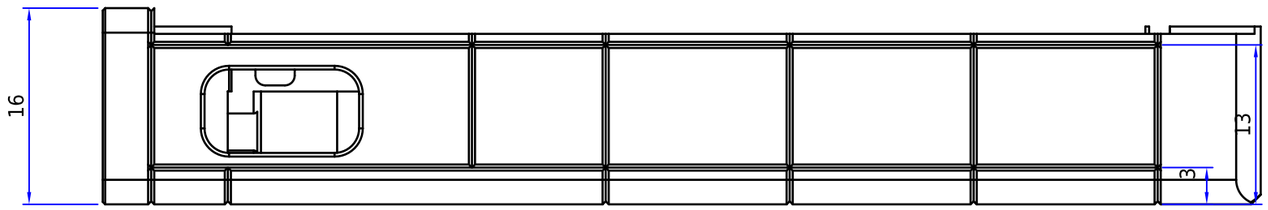
1 Material: aluminium 6061-T6. Geometry per FlashKa_tray.step; this sheet adds threads, fits, finish and notes.
2 General tolerances ISO 2768-mK; toleranced values are critical.
3 Threads are tapped before anodising and gauged after finishing.
4 Break all sharp edges 0.1-0.2. Deburr the USB-C and S2 openings and the floor holes from inside and the window bore edge: glass (10.0 x 1.0) and DTP board (10.0 +/-0.1) must drop in.
5 Finish: fine glass-bead blast, then anodise type II matte black (8-12 um). Tolerances apply after finishing.
6 Window bore floor (X 5.35) is the LED board contact face: flat 0.03, Ra 1.6, no burr or anodise step.
7 Thin walls in the STEP (mm): 1.075 window bore to control face; 1.1 laser bore to +Y face; 0.875 front tap to head back face; 1.025 front tap to window bore and bore floor; 1.0 tail front wall; 0.975 floor between the main counterbore and a slide hole. Finish them in a light last pass.
8 Inner vertical corners R2.0 (pocket 13.0 deep); the bay corner reliefs (12) are intended. Floor edges stay sharp (flat end mill).
9 Sections: A-A at Y -3.825 (window axis), D-D at Y 5.85 (laser axis), B-B at X 30, C-C at X 7.0 (front taps).

MAIN BODY (TRAY)

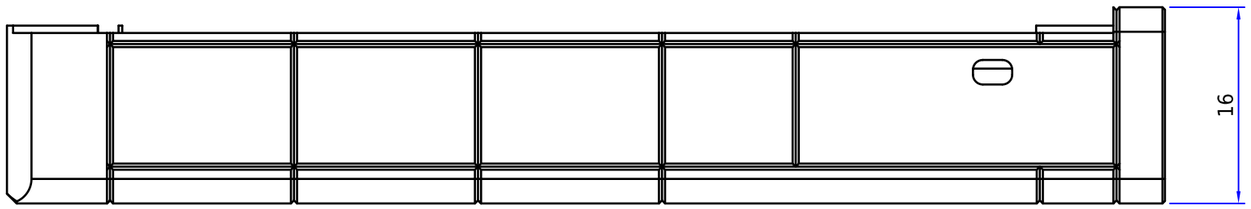
Part: FK-H-01 FlashKa_tray.step	SCALE 2:1 (details as marked) SHEET 1/2
FlashKa Keychain v0.6 2026-10-16	Units mm | ISO 2768-mK

BOTTOM VIEW (from -Z) items 9, 10, 14-21



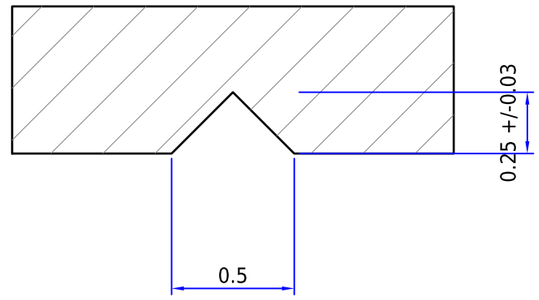


SIDE VIEW from -Y items 14, 15, 16



SIDE VIEW from +Y items 14, 15, 16

DETAIL V (seam groove, 90 deg, 40:1) items 14-16



ID	FEATURE	SPECIFICATION	TOLERANCE / NOTE
14	V-ring seam (lid joint)	90 deg V groove, 0.25 deep, round the body at X 3.95 (apex); lid and tray meet in a rabbet (lip 2.1)	depth +/-0.03
15	Frame seams 6x	90 deg V, 0.25 deep, rings round the body at X = 10.2, 30.1, 41, 56, 71, 86; continuous over the lid joint; A, B in halves	depth +/-0.03 X +/-0.1
16	Stringer seams	90 deg V, 0.25 deep: sides Z 3.0 and 13.0 (X 3.95 to 86.0); lid and bottom at Y +/- 3.6 (X 41.0 to 86.0)	depth +/-0.03
17	Seam clearances	metal under any groove >= 1.1; seams >= 0.5 from any opening or hole; numbers: housing_interface.json	
18	Tail window (bottom)	5.2 x 14.0, 1.0 deep, corners R0.8 / R1.8, chamfer C0.3; X 88.0 to 93.2	+/-0.1
19	Lanyard slot, THRU	stadium 4.4 x 7.6, centre X 90.2; chamfer C0.35 both faces	+/-0.1
20	Tail facets	2.0 x 45 deg on the four tail corners; end edge C0.8	ISO 2768-mK
21	Laser marking (artwork)	FlashKa_marking_bottom / _minus_y / _plus_y .dxf: layer MARK = laser, layer REF = outline (do not mark)	

NOTES
1 Seams are 90 deg V grooves cut with a ball-free V cutter (tip R <= 0.03) after the final milling and before blasting. Do not deepen them: the metal under them is 1.1 at the thinnest spot.
2 The bottom view is drawn with all seams; marking artwork is laser-marked after anodising (black anodise type II removed by the laser, bare aluminium shows).
3 Bottom-face features (9), (10) are on sheet 1 (items) and here (positions). Floor edges of the holes get the 0.2 x 45 deg chamfer on the outside.
4 Finish: glass-bead blast, anodise type II matte black (8-12 um), then laser marking.

MAIN BODY (TRAY) - EXTERIOR

Part: FK-H-01 FlashKa_tray.step	SCALE 2:1 (details as marked) SHEET 2/2
FlashKa Keychain v0.6 2026-10-16	Units mm | ISO 2768-mK